Autodesk® Scaleform®

Scaleform

DirectX 9 Scaleform 3D

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3.09

2013 10 7

Autodesk® GAMEWARE

Scaleform 4.3

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1

Scaleform UI
Adobe Flash Studio GPU
Scaleform Flash Studio

UI Scaleform 3D Flash Flash
3D 3D Flash UI Scaleform UI

Scaleform Flash
DirectX ShadowVolume SDK

Scaleform Scaleform
Scaleform
Scaleform

Scaleform Scaleform 4.0

DirectX
SDK ShadowVolume Scaleform “cannot create
renderer” Scaleform Player

2

Scaleform Scaleform
https://gameware.autodesk.com/scaleform/developer/

- Web Scaleform SDK
- PDF
- Scaleform C++ API

- XML Scaleform XML
- Scale9 Grid Scale9Grid
- IME Scaleform IME Flash IME
- Scaleform

3

3.1

Windows Scaleform

Scaleform installer DirectX Scaleform C:\Program Files\Scaleform\GFx SDK 4.4

3rdParty

Scaleform libjpeg zlib

Projects

Visual Studio

App\Samples

Apps\Tutorial

Bin

Flash

FxPlayer: Flash HUD

IME: IME Flash

Samples: Flash FLA

Video Demo Scaleform

Win32: Scaleform Player

AMP: AMP Flash

GFxExport: GFxExport Flash

7

Include

Scaleform

Lib

Scaleform

Resources

CLIK

Doc

2 PDF

3.2

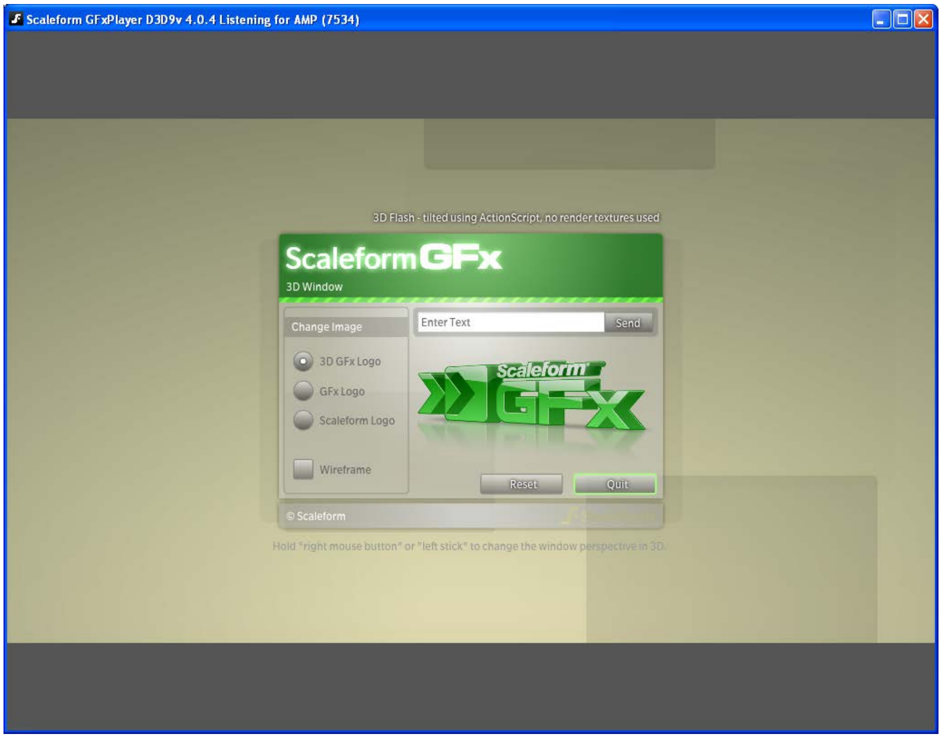
Scaleform C:\Program Files (x86)\Scaleform\GFx SDK 4.4\Projects\Win32\Msvc90\Demos\GFx 4.4 Demos.sln Demos.sln Visual Studio .sln D3D9_Debug_Static Scaleform Player

GFx SDK 4.4\Bin\Win32\Msvc90\GFxPlayer\GFxPlayer_D3D9_Debug_Static.exe

Start → All Programs → Scaleform → GFx SDK 4.4 → Demo GFx Player D3Dx SWF Scaleform Flash

3.3 Scaleform Player SWF

Start → All Programs → Scaleform → GFx SDK 4.4 → GFx Players→ GFx Player D3D9 C:\Program Files\Scaleform\GFx SDK 4.4\Bin\Data\AS2\Samples\SWFToTexture 3DWindow.swf



1 Flash

F1

1. CTRL+F FPS
2. CTRL+W Scaleform Flash
CTRL+W
3. CTRL
4. CTRL
5. CTRL+Z
6. CTRL+U
7. F2

Flash
800x600 1600x1200

Scaleform "Scaleform" Flash
Scaleform

"3D Scaleform Logo" CTRL+W
CTRL+A EdgeAA

FSAA

AA Scaleform EdgeAA

DP

EdgeAA

Scaleform Player Flash CPU CPU

Scaleform CTRL+Y

60 CPU

release debug

Scaleform debug Scaleform

3.4

Scaleform 4.4 Scaleform\GFx SDK 4.4\Apps\Tutorial SLN

Scaleform→ GFx SDK 4.4→ Tutorial Windows for Visual Studio 2005 Visual Studio 2008 “Debug”

2 ShadowVolume

3.5 Scaleform

Scaleform Visual Studio Tutorial
$(GFXSDK) SDK
C:\Program Files\Scaleform\GFx SDK 4.4\
$(GFXSDK) “Msvc90 ”
Visual Studio 2008 Visual Studio 2010 Msvc90 Msvc10

Scaleform
$(GFXSDK)\Src
$(GFXSDK)\Include
Visual Studio “Additional Include Directories”

AS3 AS3 “GFx/AS3/AS3_Global.h”
AS3 Scaleform LITE

The following library directories should be added to the “Additional Library Directories” field for all build configurations:

"$(DXSDK_DIR)\Lib\x86";
"$(GFXSDK)\ 3rdParty\expat-2.1.0\lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK) \3rdParty\pcre\Lib\$(PlatformName)\Msvc90\Release”;
"$(GFXSDK)\3rdParty\zlib-1.2.7\Lib\$(PlatformName)\Msvc90\Release";
"$(GFXSDK)\3rdParty\libpng-1.5.13\Lib\$(PlatformName)\Msvc90\Release”;
"$(GFXSDK)\3rdParty\curl-7.29.0\lib\$(PlatformName)\Msvc90\Release
"$(GFXSDK)\3rdParty\jpeg-8d\Lib\$(PlatformName)\Msvc90\Debug;"
"$(GFXSDK)\Lib\$(PlatformName)\Msvc90\$(ConfigurationName)

Msvc90 Visual Studio

Scaleform
libgfx.lib
libgfx_as2.lib
libgfx_as3.lib
libgfx_air.lib
libgfxexpat.lib
libjpeg.lib
zlib.lib
libcurl.lib

libpng.lib
libgfxrender_d3d9.lib
libgfxsound_fmod.lib

.vcproj
Scaleform include Tutorial\Section3.5

3.5.1 AS3_Global.h and Obj\AS3_Obj_Global.xxx

AS3_Global.h Obj\AS3_Obj_global.xxx

AS3_Obj_Global.xxx “ ”ActionScript 3 swf
“ ”(script) GlobalObjectCPP C++
Scaleform VM

AS3_Global.h ClassRegistrationTable
AS3 C++
ClassRegistrationTable
AS3_Global.h

ClassRegistrationTable include
ClassRegistrationTable AS3_Global.h

AS3 VM
“ ”

4

Scaleform 3D Unreal® Engine 3, Gamebryo™, Bigworld®, Hero Engine™, Touchdown Jupiter Engine™, CryENGINE™, Trinigy Vision™ Scaleform

Scaleform DirectX DirectX ShadowVolume SDK DirectX DXUT 2D 3D

Scaleform Flash Scaleform DXUT

DXUT
Win32 DirectX Scaleform SDK GFxPlayerTiny.cpp

4.1 Flash

Flash 3D GFx::Loader Flash GFx::MovieDef Flash GFx::Movie Render::Renderer2D Render::HAL DirectX Scaleform API
/

Scaleform

```
//                   GFx::Loader
Loader            gfxLoader;

//    SWF/GFx           GFx::MovieDef
Ptr<MovieDef>   pUIMovieDef;

//                      GFx::Movie
Ptr<Movie>  pUIMovie;

// Renderer
Ptr<Render::D3D9::HAL>  pRenderHAL;
Ptr<Render::Renderer2D> pRenderer;
MovieDisplayHandle      hMovieDisplay;
```

#2 GFx::System

Scaleform [GFx::System](#) Scaleform WinMain

```
//                   GFx::System
GFx::System           gfxInit;
```

GFx::System Scaleform Scaleform WinMain GFx::System Scaleform Tutorial

GFx::System

GFx::System GFxTutorial GFx::System

GFx::System

GFx::System::Init() GFx::System::Destroy()

#3:

Scaleform WinMain InitApp()
InitApp()

```
gfx = new GFxTutorial();
assert(gfx != NULL);
```

```
if(!gfx->InitGFx())
      assert(0);
```

GFxTutorial GFx::Loader GFx::Loader
SWF/GFx SWF/GFx
SWF/GFx GFx::Loader
GFx::Log,

GFxTutorial::InitGFx() GFx::Loader GFx::Loader
SetLog Scaleform
Scaleform PlayerLog
GFx::Log

```
//          —Scaleform
//
gfxLoader->SetLog(Ptr<Log>(*new GFxPlayerLog()));
```

GFx::Loader GFx::FileOpener
GFx::FileOpener

```
//
Ptr<FileOpener> pfileOpener = *new FileOpener;
gfxLoader->SetFileOpener(pfileOpener);
```

Render::HAL Scaleform D3D9
D3D Scaleform
HAL::InitHAL Render HAL IDirect3DDevice9
Scaleform DX9 UI

```
pRenderHAL = *new Render::D3D9::HAL();
if (!(pRenderer = *new Render::Renderer2D(pRenderHAL.GetPtr())))
     return false;
```

Scaleform Render::HAL Render::D3D9::HAL
Scaleform

#4 Flash

GFx::Loader GFx::MovieDef

GFx::MovieDef

```
//
pUIMovieDef = *gfxLoader.CreateMovie(UIMOVIE_FILENAME,
                                        Loader::LoadKeepBindData |
                                        Loader::LoadWaitFrame1, 0);
```

LoadKeepBindData D3D

LoadWaitFrame1 CreateMovie

GFx::ThreadedTaskManager

Memory System Overview

#5

GFx::MovieDef GFx::Movie GFx::Movie

```
pUIMovie = *pUIMovieDef->CreateInstance(true, 0, NULL);
assert(pUIMovie.getPtr() != NULL);
```

CreateInstance false

Flash

Memory System

Overview

Advance() false CreateInstance

```
//
pUIMovie->Advance(0.0f, 0, true);

//
MovieLastTime = timeGetTime();
```

Advance

3D

```
pUIMovie->SetBackgroundAlpha(0.0f);
```

Flash 3D

#6

Scaleform 时必须从 Render::D3D9::HAL 获取 DirectX 设备的句柄和参数描述信息。在 D3D 设备创建和 Scaleform 执行渲染前需要调用 Render::D3D9::HAL::InitHAL 。 / D3D InitHAL

ShadowVolume OnResetDevice DXUT GFxTutorial OnResetDevice

```
pRenderHAL->InitHAL(
    Render::D3D9::HALInitParams(pd3dDevice, presentParams,
                                Render::D3D9::HALConfig_NoSceneCalls));
```

InitHAL()**函数的**调用传递 D3D 设备和参数描述信息到 Scaleform。 HAL _NoSceneCalls Scaleform DirectX BeginScene() EndScene() ShadowVolume OnFrameRender

#7

D3D D3D ShadowVolume OnLostDevice Render::HAL GFxTutorial OnLostDevice D3D

```
pRenderHAL->ShutdownHAL();
```

DXUT Win32/DirectX Scaleform SDK GFxPlayerTiny.cpp

#8

Scaleform GFxTutorial

WinMain GFxTutorial

```
delete gfx;
gfx = NULL;
```

DirectX 9 Scaleform

InitHAL() ShutdownHAL()

DirectX 9 InitHAL D3DPOOL_DEFAULT

InitHAL D3DPOOL_DEFAULT

ShutdownHAL D3DPOOL_DEFAULT DXUT

ShadowVolume DXUT OnResetDevice D3DPOOL_DEFAULT

OnLostDevice GFxTutorial::OnLostDevice ShutdownHAL

GFxTutorial::OnResetDevice InitHAL

InitHAL ShutdownHAL

D3DPOOL_DEFAULT

#9

D3D

GFxTutorial::OnResetDevice

```
//
RECT windowRect = DXUTGetWindowClientRect();
DWORD windowWidth = windowRect.right - windowRect.left;
DWORD windowHeight = windowRect.bottom - windowRect.top;
pUIMovie->SetViewport(windowWidth, windowHeight, 0, 0,
                      windowWidth, windowHeight);
```

SetViewport PC

Scaleform

OpenGL

#10 DirectX

ShadowVolume OnFrameRender() D3D
BeginScene() EndScene() EndScene()
GFxTutorial::AdvanceAndRender()

```
void AdvanceAndRender(void)
{
        DWORD mtime = timeGetTime();
        float deltaTime = ((float)(mtime - MovieLastTime)) / 1000.0f;
        MovieLastTime = mtime;

        pUIMovie->Advance(deltaTime, 0);
        pRenderer->BeginFrame();

        if (hMovieDisplay.NextCapture(pRenderer->GetContextNotify()))
        {
            pRenderer->Display(hMovieDisplay);
        }

        pRenderer->EndFrame();
}
```

Advance() deltaTime
GFx::Movie::Advance

#11

[Render::Renderer2D::Display](#) DirectX D3D
D3D
Render::Renderer2D::Display D3D

Scaleform
DX9

DX9 GFxTutorial::AdvanceAndRender()

```
// Scaleform DirectX
g_pStateBlock->Capture();

//
gfx->AdvanceAndRender();

// DirectX
g_pStateBlock->Apply();
```

#12 UI

DXUT 4.1
ShadowVolume.cpp DXUT

```
// Disable default UI
...
```

DirectX Flash

3 Scaleform Flash ShadowVolum

4.2

GFx::Movie::SetViewport Flash
Flash Scaleform

•

•

4:3 Scaleform

800x600 1600x1200

Scaleform

Scaleform

GFx::Movie::SetViewScaleMode

GFxTutorial::InitGFx() GFx::Movie

```
pUIMovie->SetViewScaleMode(Movie::SM_ShowAll);
```

SetViewScaleMode

SM_NoScale	Flash
SM_ShowAll	
SM_ExactFit	
SM_NoBorder	

SetViewAlignment SM_NoScale

SM_ShowAll SetViewAlignment

```
pUIMovie->SetViewAlignment(Movie::Align_CenterRight);
```

SetViewScaleMode SetViewAlignment

SetViewAlignment SetViewScaleMode SM_NoScale

Scaleform

d3d9guide.fla

ActionScript C++ SetViewScaleMode

SetViewAlignment ActionScript Stage (State.scaleMode, Stage.align)

4.3

ShadowVolume Scaleform Flash Flash

GFx::Movie::HandleEvent GFx::Event

GFx::Movie

4.3.1

ShadowVolume MsgProc Win32 GFxTutorial::ProcessEvent

Scaleform WM_MOUSEMOVE,

WM_LBUTTONDOWN WM_LBUTTONUP

```
void ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam, LPARAM lParam,
                  bool *pbNoFurtherProcessing)
{
      int mx = LOWORD(lParam), my = HIWORD(lParam);
      if(pUIMovie)
      {
            if(uMsg == WM_MOUSEMOVE)
            {
                  MouseEvent mevent(GFx::Event::MouseMove, 0, mx, my);
                  pUIMovie->HandleEvent(mevent);
            }
            else if(pMovieButton && uMsg == WM_LBUTTONDOWN)
            {
                  ::SetCapture(hWnd);
                  MouseEvent mevent(GFx::Event::MouseDown, 0, mx, my);
                  pUIMovie->HandleEvent(mevent);
            }
            else if(pMovieButton && uMsg == WM_LBUTTONUP)
            {
                  ::ReleaseCapture();
                  MouseEvent mevent(GFx::Event::MouseUp, 0, mx, my);
                  pUIMovie->HandleEvent(mevent);
            }
      }
}
```

Scaleform

#1:

```
pMovie->SetViewport(screen_width, screen_height, 0, 0, screen_width,
                                                 screen_height, 0);
```

(0, 0)

Scaleform

#2

```
pMovie->SetViewport(screen_width, screen_height, 0, 0, screen_width / 4,
                                          screen_height / 4, 0);
```

HandleEvent

Scaleform

#3

```
movie_width = screen_width / 6;
movie_height = screen_height / 6;
pMovie->SetViewport(screen_width, screen_height,
                         screen_width / 2 – movie_width / 2,
                         screen_height / 2 – movie_height / 2,
                         movie_width, movie_height);
```

(0,0) (screen_width / 2 – movie_width / 2, screen_height / 2 – movie_height / 2)

Flash GFx::Movie::SetViewAlignment GFx::Movie::SetViewport SetViewAlignment SetViewScaleMode Scaleform

4.3.2

GFx::Movie:HandleEvent GFx::KeyEvent GFx::CharEvent:

```
KeyEvent(EventType eventType = None,
                  Key::Code code = Key::None,
                  UByte asciiCode = 0,
                  UInt32 wcharCode = 0,
                  UInt8 keyboardIndex = 0)

CharEvent(UInt32 wcharCode, UInt8 keyboardIndex = 0)
```

GFx::KeyEvent GFx::CharEvent ASCII Windows GFx::CharEvent WM_CHAR GFx::KeyEvents WM_SYSKEYDOWN, WM_SYSKEYUP, WM_KEYDOWN WM_KEYUP

- 'c' SHIFT : GFx::KeyEvent WM_KEYDOWN
  - 'c'
  -
  - SHIFT
- GFx::CharEvent WM_CHAR " " ASCII 'C' GFx
- 'c' GFx::KeyEvent WM_KEYUP GFx::CharEvent
- F5 GFx::KeyEvent F5 ASCII GFx::CharEvent

GFx::KeyEvent GFx_Event.h Flash GFxPlayerTiny.cpp Scaleform Player Windows Flash ProcessKeyEvent 3D

```
void ProcessKeyEvent(Movie *pMovie, unsigned uMsg, WPARAM wParam, LPARAM
lParam)
```

Windows WndProc WM_CHAR, WM_SYSKEYDOWN, WM_SYSKEYUP, WM_KEYDOWN WM_KEYUP Scaleform pMovie

GFx::KeyEvent GFx::CharEvent GFx::CharEvents ASCII Unicode IME . IME IME GFx::KeyEvent Page Up Page Down

4.3

3D “Settings” DX9 C++ d3d9guide.fla

Flash UI UI Scaleform DXUT DXUT Scaleform Scaleform DXUT Scaleform Flash UI

“Change Mesh” “Change Mesh” Flash

4.3.3

3D

UI 3D GFx::Movie::HitTest Flash GFxTutorial::ProcessEvent [HitTest] UI DXUT

```
bool processedMouseEvent = false;
if(uMsg == WM_MOUSEMOVE)
{
      MouseEvent mevent(GFx::Event::MouseMove, 0, (float)mx, (float)my);
      pUIMovie->HandleEvent(mevent);
      processedMouseEvent = true;
}
else if(uMsg == WM_LBUTTONDOWN)
{
      ::SetCapture(hWnd);
      MouseEvent mevent(GFx::Event::MouseDown, 0, (float)mx, (float)my);
      pUIMovie->HandleEvent(mevent);
      processedMouseEvent = true;
}
else if(uMsg == WM_LBUTTONUP)
{
```

```
            ::ReleaseCapture();
            MouseEvent mevent(GFx::Event::MouseUp, 0, (float)mx, (float)my);
            pUIMovie->HandleEvent(mevent);
            processedMouseEvent = true;
      }

      if(processedMouseEvent && pUIMovie->HitTest((float)mx, (float)my,
            Movie::HitTest_Shapes))
            *pbNoFurtherProcessing = true;
```

4.3.4

"Change Mesh"

4.3.2 W,S,A,D Q 3D

Scaleform 3D

6.1.2

C++

4.3.5

Windows API Scaleform

Windows 7

WINVER=0x601 " " Windows API

Windows 7 ShadowVolume.cpp `#define ENABLE_MULTITOUCH`

`#include <windows.h>`

RegisterTouchWindow RegisterTouchWindow MsgProc

WM_TOUCH

```
if(uMsg == WM_TOUCH)
{
      ProcessTouchEvent(pUIMovie, uMsg, wParam, lParam);
}
```

ProcessTouchEvent WM_TOUCH Windows

GFx::TouchEvent HandleEvent GFx::TouchEvent

ID OS x/y 0..1

```
          TouchEvent( EventType evtType,
                      unsigned id,
                      float _x,
                      float _y,
                      float wcontact = 0,
                      float hcontact = 0,
                      bool primary = true,
                      float pressure = 1.0f)
```

SWF GFx 0

MultitouchInterface

GFx::GestureEvent

Scaleform

```
class FxPlayerMultitouchInterface : public MultitouchInterface
{
public:
  //
    virtual unsigned GetMaxTouchPoints() const { return 2; }

  //
    virtual UInt32   GetSupportedGesturesMask() const { return 0; }

  //
    virtual bool     SetMultitouchInputMode(MultitouchInputMode) { return
true; }
};
```

SetMultitouchInterface

5

5.1

Scaleform

Flash

TTF

Windows Linux Scaleform

https://gameware.autodesk.com/scaleform/developer/?action=doc

5000

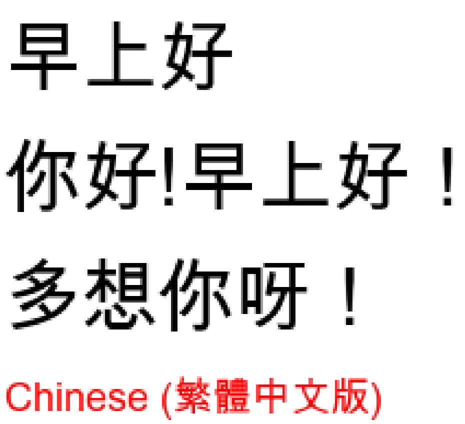
Scaleform

Scaleform

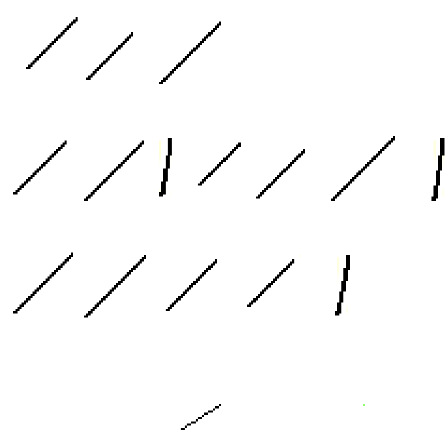
Scaleform TTF

Scaleform “ ” “ ”

Bin\Data\AS2\Samples\FontConfig sample.swf Scaleform Player D3D9



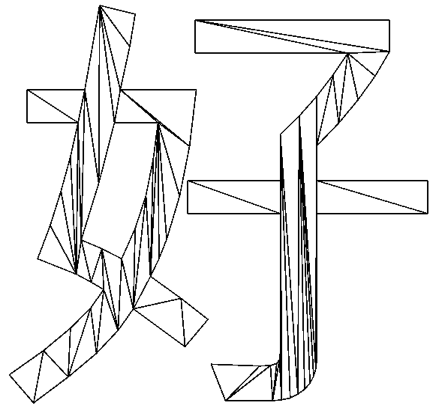
3a



4b

CTRL+W

4c



4d

4c 4d Scaleform

Flash

gfx_2.1_texteffects_sample.zip


Isotropic shadow with offset
Text Shadow
文本陰影
Anisotropic glow, BlurX=2, BlurY=7
Text Glow 1
文本煥發 1
Isotropic glow, BlurX=BlurY=4, Knockout
TEXT GLOW 2
文本煥發 2
Rotating glow
Text Glow 3
文本煥發 3
Changing blurriness
Text Glow 4
文本煥發 4

5

6 C++ Flash

Flash

Flash

Scaleform C++

Scaleform

C++

6.1 *C++*

Scaleform C++ FSCommand

ExternalInterface FSCommand ExternalInterface GFx::Loader C++

FSCommand fscommand

ExternalInterface flash.external.ExternalInterface.call

GFx::Value 6.1.2

FSCommands ExternalInterface

fscommands gfxexport SWF

fscommands –fstree –fslist –fsparams ExternalInterface

fscommands

6.1.1 FSCommand

fscommand

```
fscommand("setMode", "2");
```

fscommand ExternalInterface

GFx FSCommand GFx::FSCommandHandler

FSCommand GFx::Loader GFx::Movie

GFx::Movie fscommand Scaleform PlayerTiny

(“FxPlayerFSCommandHandler”) ShadowVolume

6.1

GFx::FSCommandHandler

```
class OurFSCommandHandler : public FSCommandHandler
{
    public:
    virtual void Callback(Movie* pmovie,
                           const char* pcommand, const char* parg)
    {
        GFxPrintf("FSCommand: %s, Args: %s", pcommand, parg);
    }
};
```

Callback fscommand fscommand

GFxTutorial::InitGFx() GFx::Loader

```
//      FSCommand
Ptr<FSCommandHandler> pcommandHandler = *new OurFSCommandHandler;
gfxLoader->SetFSCommandHandler(pcommandHandler);
```

GFx::Loader GFx::Movie SetFSCommandHandler

fscommand ShadowVolume
“Toggle Fullscreen”

```
FSCommand: ToggleFullscreen, Args:
```

Open Flash/HUDMgr.as in Flash Studio. This class is referenced from d3d9guideAS3.fla. The association between this external class and the Flash content is made by setting the ActionScript class for the “hudMgr” Symbol in d3d9guideAS3’s Symbol Library.

Compare the events printed to the screen with the fscommand() calls made from the HUDMgr class in the following function:

```
function toggleFullscreen(ev:MouseEvent) {
                fscommand("ToggleFullscreen");
}
```

As an exercise, change fscommand(“ToggleFullscreen”) to fscommand(“ToggleFullscreen”, String(hud.visible)) to print the value of the hud.visible value to C++. Export the flash movie (CTRL+ALT+Shift+S) and replace d3d9guideAS3.swf.

fscommand()

FSCommand ShadowVolume

fscommand

```
if(strcmp(pcommand, "ToggleFullscreen") == 0)
    doToggleFullscreen = true;
```

DXUT OnFrameMove OnFrameMove

```
if(doToggleFullscreen)
{
    doToggleFullscreen = false;
    DXUTToggleFullScreen();
}
```

fscommand doToggleFullscreen DXUT OnFrameMove

6.1.2 ExternalInterface

Flash ExternalInterface fscommand

[ExternalInterface](#) fscommand

```
class OurExternalInterfaceHandler : public ExternalInterface
{
  public:
    virtual void Callback(Movie* pmovieView,
                          const char* methodName,
                          const Value* args,
                          unsigned argCount)
    {
        GFxPrintf("ExternalInterface: %s, %d args: ",
                                      methodName, argCount);
        for(unsigned i = 0; i < argCount; i++)
        {
            switch(args[i].GetType())
```

```
                        {
                          case Value::VT_Null:
                              GFxPrintf("NULL");
                              break;
                          case Value::VT_Boolean:
                              GFxPrintf("%s", args[i].GetBool()? "true" : "false");
                              break;
                          case Value::VT_Int:
                              GFxPrintf("%s", args[i].GetInt());
                              break;
                          case Value::VT_Number:
                              GFxPrintf("%3.3f", args[i].GetNumber());
                              break;
                          case Value::VT_String:
                              GFxPrintf("%s", args[i].GetString());
                              break;
                          default:;
                              GFxPrintf("unknown");
                              break;
                        }
                        GFxPrintf("%s", (i == argCount - 1) ? "" : ", ");
                  }
                  GFxPrintf("\n");
            }
      };
```

GFx::Loader

```
      Ptr<ExternalInterface> pEIHandler = *new OurExternalInterfaceHandler;
      gfxLoader.SetExternalInterface(pEIHandler);
```

Open d3d9HUD.as used by d3d9guideAS3.fla and look at the ActionScript for the following functions:

```
      function onSubmit(path:String) {
            ExternalInterface.call("MeshPath", path);
            CloseMeshPath();
      }

      function onFocusIn(ev: FocusHandlerEvent) {
            ExternalInterface.call("MeshPathFocus", true);
      }

      function onFocusOut(ev: FocusHandlerEvent) {
            ExternalInterface.call("MeshPathFocus", false);
      }
```

MeshPathFocus

(true false) “MeshPathFocus”

```
Callback! MeshPathFocus, nargs = 1
  arg(0) = true

Callback! MeshPathFocus, nargs = 1
   arg(0) = false
```

GFxTutorial

```
if(strcmp(methodName, "MeshPathFocus") == 0 && argCount == 1 &&
args[0].GetType() == Value::VT_Boolean)    {
      if (args[0].GetType() == Value::VT_Boolean)
            gfx->SetTextboxFocus(args[0].GetBool());
}
```

GFxTutorial::ProcessEvent

3D

```
if(uMsg == WM_SYSKEYDOWN || uMsg == WM_SYSKEYUP ||
   uMsg == WM_KEYDOWN    || uMsg == WM_KEYUP    ||
   uMsg == WM_CHAR)
{
      if(textboxHasFocus || wParam == 32 || wParam == 9)
      {
            ProcessKeyEvent(pUIMovie, uMsg, wParam, lParam);
            *pbNoFurtherProcessing = true;
      }
}
```

(ASCII 32) tab(ASCII 9) “Toggle UI” “Settings”

“Change Mesh”

OurExternalInterfaceHandler::Callback

```
static bool doChangeMesh = false;
static wchar_t changeMeshFilename[MAX_PATH] = L"";
```

```
...

if(strcmp(methodName, "MeshPath") == 0 && argCount == 1)
{
	doChangeMesh = true;
	const char *filename = args[0].GetString();
	MultiByteToWideChar(CP_ACP, MB_PRECOMPOSED, filename, -1,
		changeMeshFilename, _countof(changeMeshFilename));
}
```

DXUT OnFrameMove DXUT

6.2 C++

6.1 C++ Scaleform
C++ Scaleform C++

6.2.1

Scaleform GetVariable SetVariable 6.2
HUD (fxplayer.swf) Scaleform Player F5

```
void GFxTutorial::ProcessEvent(HWND hWnd, unsigned uMsg, WPARAM wParam,
                               LPARAM lParam, bool *pbNoFurtherProcessing)
{
	int mx = LOWORD(lParam), my = HIWORD(lParam);

	if(pHUDMovie && uMsg == WM_KEYDOWN)
	{
		if(wParam == VK_F5)
		{
			int counter = (int)pHUDMovie-
			                  >GetVariableDouble("_root.counter");
			counter++;
			pHUDMovie->SetVariable("_root.counter",
			                  Value((double)counter));
			char str[256];
			sprintf_s(str, "testing! counter = %d", counter);
			pHUDMovie->SetVariable("_root.MessageText.text", str);
```

```
        }
    }
    ...
```

GetVariableDouble _root.counter GetVariableDouble
SetVariable _root.counter
GFx::Movie GetVariable SetVariable

fxplayer Flash MessageText
HUDText _root.counter SetVariable

Flash TextField.text GFx::Movie::Invoke
6.2.2
Scaleform

8 SetVariable HUD

GFx::Value

SetVariable GFx::Movie::SetVarType "
"

_root.mytextfield 3 SetVariable("_root.mytextfield.text", "testing",

SV_Normal） 1 SV_Sticky
3 _root.mytextfield.text C++

SetVariableArray Flash
GFxTutorial::InitGFx() _root.SceneData

```
//
Value sceneData[3];
sceneData[0].SetString("Scene with shadow");
sceneData[1].SetString("Show shadow volume");
sceneData[2].SetString("Shadow volume complexity");
pUIMovie->SetVariableArraySize("_root.SceneData", 3);
pUIMovie->SetVariableArray(Movie::SA_Value,
      "_root.SceneData", 0, sceneData, 3);
```

6.1
ShadowVolume

C++
C++

6.2.2

GFx::Movie::Invoke
UI

GFx::Movie::Invoke “Change Mesh” F6 F7
SetVariable
SetVariable Invoke ActionScript

GFx::Movie::Invoke WM_CHAR openMeshPath

```
...

else if (wParam == VK_F6)
{
      bool retval = pHUDMovie->Invoke("root.ui.hud.OpenMeshPath", "");
      GFxPrintf("root.ui.hud.OpenMeshPath returns '%d'\n", (int) retval);
}
```

```
else if (wParam == VK_F7)
{
      const char *retval = pHUDMovie->Invoke(
                              "root.ui.hud.CloseMeshPath", "");
      GFxPrintf("root.ui.hud.CloseMeshPath returns '%d'\n", (int) retval);
}
```

Scaleform

GFx::Movie::Advance GFx::MovieDef::CreateInstance

initFirstFrame true 1

printf GFx::Value

InvokeArgs va_list Invoke

InvokeArgs printf vprintf

6.3 *Flash*

C++ SWF

C++

MMOG HUD HUD

HUD

HUD

SWF C++

GFx::Movie

loadMovie unloadMovie

SWF GFx::Movie SWF GFx::Movie

C++

ActionScript container.swf

container.swf ActionScripts

MovieClipLoader

```
var mclLoader:MovieClipLoader = new MovieClipLoader();
```

ActionScript Flash

```
function loadMovie(url:String):void {
        var loader = new Loader();
        loader.load( new URLRequest( url ) );
        addChild( loader )
    }
```

`loadMovie`

root.inventoryWindow.widget1.counter

Flash root root

level 6 root.counter level6.counter

`loadMovie` root.myMovie root.counter

lockroot = true Lockroot ActionScript

root root root ActionScript

Adobe Flash

GFx::Movie

Container.fla

`loadMovie` loadClip

Loader Container.fla

ActionScript ExternalInterface C++

```
function loadMovie(url:String):void {
        var loader = new Loader();
        loader.load( new URLRequest( url ) );
        loader.contentLoaderInfo.addEventListener( Event.OPEN, handleLoadOpen,
                                                   false, 0, true );
        loader.contentLoaderInfo.addEventListener( Event.INIT, handleLoadInit,
                                                   false, 0, true );
        loader.contentLoaderInfo.addEventListener( Event.PROGRESS,
                                          handleLoadProgress, false, 0, true );
        loader.contentLoaderInfo.addEventListener( Event.COMPLETE,
```

```
                                      handleLoadComplete, false, 0, true );
    addChild( loader )
}

function handleLoadOpen( e:Event ):void {
    trace("Event.OPEN - Load Started!");
}
function handleLoadInit( e:Event ):void {
    trace("Event.INIT - Loaded Content Ready!");
}
function handleLoadProgress( e:Event ):void {
    trace("Event.PROGRESS - Load Progress!");
}
function handleLoadComplete( e:Event ):void {
    trace("Event.COMPLETE - Load Complete!");
}
```

7

3D

“ ” Flash Studio SWF SWF

UI

UI

1. cell.x

(cellwall.jpg)

UI

cellwallRT.jpg MeshMaterialList

```
Material {
1.000000;1.000000;1.000000;1.000000;;
40.000000;
1.000000;1.000000;1.000000;;
0.000000;0.000000;0.000000;;
```

```
TextureFilename {
"cellwallRT.jpg";
}
```

2. cell.x DXUT

DXUT

CreateTexture (Render

Texture) DXSDK

DXUT

```
pd3dDevice->CreateTexture(rtw,rth,0,
D3DUSAGE_RENDERTARGET|D3DUSAGE_AUTOGENMIPMAP, D3DFMT_A8R8G8B8,
            D3DPOOL_DEFAULT, &g_pRenderTex, 0);

g_Background[0].m_pTextures[BACK_WALL] = g_pRenderTex;
ptex->Release();
```

3. AdvanceAndRender GFx

a.

```
pd3dDevice->GetRenderTarget(0, &poldSurface);
pd3dDevice->GetDepthStencilSurface(&poldDepthSurface);
```

b.

```
ptex->GetSurfaceLevel(0, &psurface);
HRESULThr = pd3dDevice->SetRenderTarget(0, psurface );
```

c. Display a

8 OpenGL

OpenGL/GLUT (Tutorial\OpenGL) Nvidia (Tessellation)
(http://developer.download.nvidia.com/SDK/10/opengl/samples.html#tessellation)
D3D9 tessellation.cpp
tessellation_original.cpp NVIDIA
Init() ShutDown AdvanceAndDisplay() GFx GFx
GFxPlayerImpl (GFxPlayerGL.cpp)

9 GFxExport

SWF SWF
SWF SWF
SWF

GFxExport SWF
DDS DXT

GFxExport

Scaleform GFxExport
d3d9guide.swf fxplayer.swf Scaleform

```
gfxexport -i DDS -c d3d9guideAS3.swf
gfxexport -i DDS -c fxplayer.swf
```

gfxexport.exe C:\Program Files\Scaleform\$(GFXSDK)
convert.bat 7

-i DDS DDS DirectX DXT
DXT

-c Scaleform
DXT

-share_images

ShadowVolume 8 Scaleform

GFx::Loader::CreateMovie

10

Scaleform

- Flash Scaleform Player SWF SDK Gamebryo 3
- FAQs
- Scale9 Scale9Grid Overview
- Scaleform SWF Flash GFx::FileOpener
- GFx::ImageCreator
- GFx::Translator